

INTRODUCTION TO ETHICAL HACKING

Understanding the mindset, role, and importance of Ethical Hackers

THINK LIKE AN ATTACKER. ACT LIKE A DEFENDER.

[TERMINAL]

```
$ course_status  
> beginner-friendly  
> authorized  
security testing  
> modern threat  
landscape
```

What Is Ethical Hacking?

The foundation of this course

[DEFINITION]

- ▶ An authorized practice of testing systems, networks, applications, and infrastructure.
- ▶ The goal is to identify security weaknesses before malicious attackers exploit them.
- ▶ An Ethical Hacker works from an attacker's perspective but with permission and a defensive objective.

**FIND THE
WEAKNESS BEFORE
THE ATTACKER
DOES.**

A Simple Real-World Example

Security testing before a real incident

[PHYSICAL_SECURITY]

- ▶ Are the doors properly locked?
- ▶ Can someone enter through a window?
- ▶ Are there CCTV blind spots?
- ▶ Are the locks strong enough?

[DIGITAL_SECURITY]

- ▶ The security expert identifies weaknesses.
- ▶ Nothing is stolen or damaged.
- ▶ The owner fixes the problems before a real criminal finds them.
- ▶ Ethical Hacking follows the same defensive principle in the digital world.

Authorization Is the Core Principle

Ethical Hacking begins with permission

NO AUTHORIZATION → NOT AN AUTHORIZED SECURITY ENGAGEMENT

[IN_SCOPE]

[TERMINAL]

```
example.com  
app.example.com  
api.example.com
```

[OUT_OF_SCOPE]

[TERMINAL]

```
Internal employee systems  
Production databases  
Third-party services
```

Ethical Hacker vs. Malicious Attacker

[ETHICAL_HACKER]

- ▶ Has permission
- ▶ Works within an agreed scope
- ▶ Identifies weaknesses
- ▶ Reports findings responsibly
- ▶ Helps improve security

[MALICIOUS_ATTACKER]

- ▶ Acts without permission
- ▶ Ignores boundaries
- ▶ Exploits weaknesses
- ▶ May steal, damage, or extort
- ▶ Causes harm or gains unauthorized benefits

Think Like an Attacker. Act Like a Defender.

[ATTACKER_QUESTIONS]

- ▶ What is exposed to the internet?
- ▶ What services are running?
- ▶ Is any software outdated?
- ▶ Are credentials weak?
- ▶ Is anything misconfigured?
- ▶ Is sensitive information exposed?

How could this organization be attacked, and how can we reduce that risk?

The Role of an Ethical Hacker

Security testing is more than simply finding a vulnerability

[ROLE_01]

Vulnerability Assessment — Identify outdated software, missing patches, exposed services, and insecure configurations.

[ROLE_02]

Penetration Testing — Validate whether a weakness is realistically exploitable and understand its potential impact.

[ROLE_03]

Security Architecture Review — Evaluate authentication, access permissions, network segmentation, and cloud security.

[ROLE_04]

Incident Readiness — Help assess detection, investigation, containment, response, and recovery capabilities.

Why Ethical Hacking Matters in 2026

[AI-DRIVEN THREATS]

Faster information gathering, automation, content generation, and social engineering.

[24/7 ATTACK SURFACE]

Cloud, APIs, mobile applications, remote employees, and third-party services.

[ZERO TRUST]

Never trust automatically. Always verify access, identity, and security controls.

[BREACH IMPACT]

Financial loss, business disruption, reputation damage, and loss of customer trust.

From Reactive Security to Proactive Security

[REACTIVE]

“What do we do after an attack?”

[PROACTIVE]

“What weaknesses can we identify and fix before an attack happens?”

Key Takeaways

[ETHICAL_HACKING_IS]

- ▶ Authorized
- ▶ Structured
- ▶ Professional
- ▶ Responsible

[ULTIMATE_GOAL]

IDENTIFY WEAKNESSES



UNDERSTAND RISK



IMPROVE SECURITY

ETHICAL HACKING

is not about proving that you can break a system.

**THINK LIKE AN ATTACKER.
ACT LIKE A DEFENDER.**



LEARN

•

TEST

•

DEFEND

